

AdaptR1: Reinforcement Learning Based Adaptive Interleaved Thinking in Multi-hop Question Answering

Yuxin Wang^{1,2,*}, Jiahao Lu^{1,3,*}, Qifeng Wu^{1,*}, Shicheng Fang^{1,3},
Chuanyuan Tan⁴, Yining Zheng¹, Xuanjing Huang^{1,2}, Xipeng Qiu^{1,3}

¹Computer Science, Fudan University

²Institute of Modern Languages and Linguistics, Fudan University

³Shanghai Innovation Institute

⁴Soochow University

{wangyuxin21, 25113050083, 25213050409, 25113050022}@m.fudan.edu.cn

{ynzheng19, xjhuang, xpqiu}@fudan.edu.cn

cytan17726@stu.suda.edu.cn

Abstract

Large Language Models (LLMs) have achieved remarkable performance in complex reasoning tasks through Chain-of-Thought (CoT) prompting. However, this approach often leads to “over-thinking,” where models generate unnecessarily long reasoning traces for simple queries and incur avoidable inference cost. While recent work has explored adaptive reasoning, existing methods typically make a single query-level decision about whether to reason. This overlooks the dynamic nature of multi-step tasks, where the need for explicit reasoning varies across intermediate stages. To address this limitation, we introduce AdaptR1, a Reinforcement Learning (RL) based framework for adaptive interleaved thinking in multi-hop Question Answering (QA). Unlike previous approaches that require Supervised Fine-Tuning (SFT) for cold-start initialization, AdaptR1 uses a fully RL-based strategy with a quality-gated efficiency reward to dynamically allocate reasoning budgets at each step. Under the Graph-R1 setting, AdaptR1 reduces average think tokens by 69.71%, with a 90.35% reduction on HotpotQA, while maintaining performance comparable to or better than standard baselines. Furthermore, our analysis reveals that overthinking in multi-hop reasoning is not uniformly distributed but occurs predominantly during the initial planning stages, highlighting the effectiveness of step-wise adaptive budget allocation.

1 Introduction

Recently, large language models (LLMs)(OpenAI et al., 2024a,b) have demonstrated remarkable capabilities across a wide range of natural language understanding and generation tasks. Despite these capabilities, LLMs still struggle with

*Equal contribution.

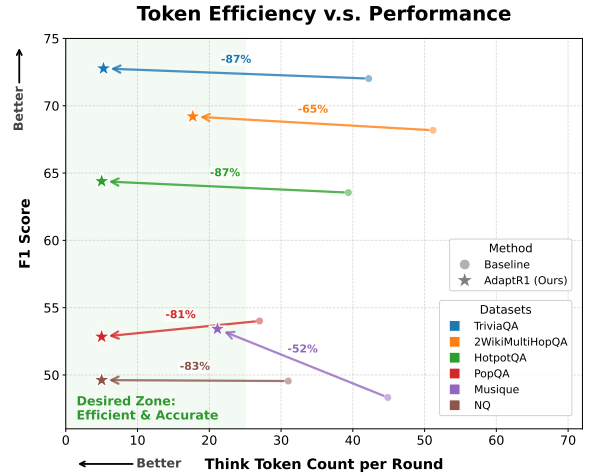


Figure 1: Comparison of F1 scores across RAG benchmarks. AdaptR1 maintains comparable performance while reducing think tokens.

tasks that require complex and multi-step reasoning, such as mathematical problem solving, logical inference, and planning. To address this limitation, researchers have explored methods to elicit stronger reasoning behavior. Relevant methods(OpenAI et al., 2024b; Shao et al., 2024) mainly include prompt-based techniques, such as Chain-of-Thought (CoT)(Wei et al., 2022) prompting, which encourages models to generate intermediate reasoning steps, and training-based approaches, including Supervised Fine-tuning (SFT)(Zheng et al., 2024) with reasoning traces and Reinforcement Learning (RL)(Shao et al., 2024), which explicitly incorporate reasoning processes into the model.

However, the introduction of CoT has also led to an emerging issue: overthinking(Kumar et al., 2025; Sui et al., 2025; Nayab et al., 2024). Instead of allocating reasoning effort proportionally to task difficulty, CoT often induces LLMs to produce unnecessarily long reasoning traces even for

AdaptR1: Adaptive thinking in each step; **High efficiency.**

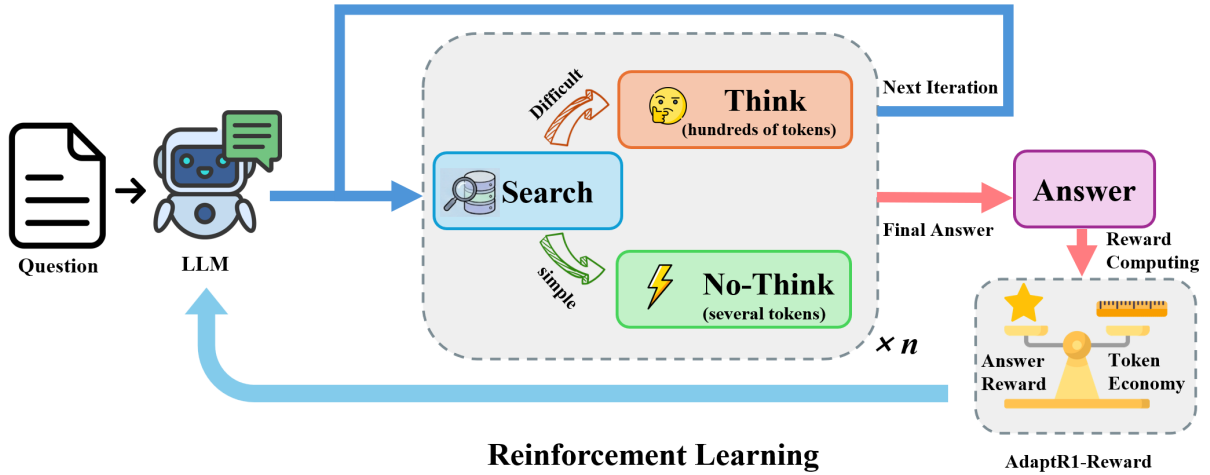


Figure 2: Framework of AdaptR1. RL teaches the model to skip explicit thinking at selected intermediate steps, reducing over-thinking and token usage.

simple queries, increasing both inference time and computational cost. To mitigate this, several research directions have emerged. For instance, SFT with preferred reasoning lengths (Chen et al., 2024; Shen et al., 2025; Luo et al., 2025c) or reinforcement learning with length-based penalties (Arora and Zanette, 2025; Team et al., 2025) encourages concise reasoning traces, while adaptive reasoning or selective thinking techniques (Zhang et al., 2025; Tu et al., 2025; Ma et al., 2025a; Chen et al., 2025b) attempt to adjust the depth of reasoning based on query complexity.

While existing methods aim to balance reasoning depth and computational efficiency, current adaptive thinking strategies typically make a single global decision—whether to think or not—on a per-query basis. However, no prior work investigates adaptive **interleaved** thinking in multi-step reasoning settings, where decisions to reason, skip, or adjust thinking effort occur dynamically across intermediate steps.

This raises a key question: *Is it necessary for an LLM to think at every step of multi-hop reasoning? If over-thinking exists within intermediate steps, how can an LLM learn to select Think or No-Think adaptively based on the difficulty of each step?*

In this paper, we introduce **AdaptR1**, an RL-based adaptive interleaved thinking method for multi-hop question answering. AdaptR1 enables adaptive reasoning at each stage of a multi-step process, allowing models to allocate reasoning effort more efficiently. Motivated by AdaptThink (Zhang et al., 2025), we use `<think>no_think</think>`

to denote skipping explicit thinking in the current step. Under the Graph-R1 setting, our method reduces average think tokens by 69.71% while maintaining answer quality. Our analysis characterizes where efficient thinking emerges across multi-hop reasoning steps and how overthinking appears within the process. Our contributions are as follows:

- We study adaptive interleaved thinking for multi-hop QA and show that over-thinking appears in intermediate reasoning steps and can be reduced through learning.
- We propose an RL-only adaptive thinking method that avoids SFT cold-start trajectories, and we design a quality-gated efficiency reward for QA tasks with continuous answer rewards such as F1.
- AdaptR1 achieves comparable or better performance in multi-hop question answering with a 69.71% average think-token reduction under the Graph-R1 setting. Extensive analysis shows that over-thinking is concentrated in the early stages of multi-hop reasoning rather than the final synthesis stage.

2 Related Works

Efficient Reasoning in LRMs. Following recent observations regarding the “over-thinking” phenomenon in long Chain-of-Thought (CoT) reasoning, adaptive thinking strategies have garnered significant attention. Existing approaches to efficiency

generally fall into two categories. The first involves intrinsic model modifications, achieved either through integrating length-based rewards in reinforcement learning (RL)(Arora and Zanette, 2025; Team et al., 2025; Aggarwal and Welleck, 2025; Hou et al., 2025; Lou et al., 2025), supervised fine-tuning (SFT) on concise responses(Chen et al., 2024; Shen et al., 2025; Luo et al., 2025c; Ma et al., 2025b; Kang et al., 2025), or by amalgamating reasoning and non-reasoning parameters (Wu et al., 2025a). The second category empowers LLMs to adaptively modulate their reasoning process based on query complexity. Prominent examples include AdaptThink (Zhang et al., 2025), AutoThink (Tu et al., 2025), HiPO (Deng et al., 2025), and ARM (Wu et al., 2025b); notably, A²FM (Chen et al., 2025b) extends this framework to encompass instant, reasoning, and agentic modes. Our research aligns with this second paradigm. However, these methods usually make a single query-level routing decision, whereas interleaved multi-hop QA requires repeated decisions after each retrieval result. This step-wise setting also makes direct comparison with single-turn pruning methods less informative, because they do not support the reason-search-answer loop evaluated here. AdaptR1 therefore addresses the unexplored challenge of mitigating over-thinking within the granular steps of multi-hop reasoning, and it learns this behavior directly through RL without SFT cold-start trajectories.

Multi-hop Question Answering. Methodologies for multi-hop Question Answering (QA) can be broadly classified into training-free and training-based paradigms. Training-free methods employ prompting strategies such as Chain-of-Thought (CoT)(Wei et al., 2022) and various retrieval-augmented frameworks including IR-CoT(Trivedi et al., 2022a), ITER-RETGEN (Shao et al., 2023), WebGPT (Nakano et al., 2021), ReAct (Yao et al., 2023), Self-RAG (Asai et al., 2024), Self-ask (Press et al., 2023), and FLARE (Jiang et al., 2023). Conversely, training-based methods—such as R1-Searcher (Song et al., 2025), DeepResearcher (Zheng et al., 2025), R3-RAG (Li et al., 2025), DeepRAG (Guan et al., 2025), SearchR1 (Jin et al., 2025a), and Graph-R1 (Luo et al., 2025a)—utilize SFT or RL to cultivate step-by-step reasoning capabilities. Despite these advancements, current literature lacks a mechanism to address over-thinking specifically within multi-hop QA contexts. Our proposed method fills this gap by

implementing an RL-exclusive adaptive thinking strategy tailored for these scenarios.

3 Preliminaries

AdaptR1 adds one adaptive instruction to the native parent prompt: at each scheduled reasoning slot, the model can either generate explicit reasoning inside `<think>...</think>` or emit `<think>no_think</think>` to skip it. In the Graph-R1 setting, tool calls use `<query>...</query>` and retrieved evidence is returned inside `<knowledge>...</knowledge>`; Search-AdaptR1 preserves Search-R1’s native `<search>/<information>` interface. The full Graph-AdaptR1 prompt is provided in Appendix A.

GRPO. Group Relative Policy Optimization (GRPO)(Shao et al., 2024) samples grouped roll-outs for each question and updates the policy with group-normalized sequence-level advantages. In AdaptR1, the standard sequence reward is replaced by the adaptive reward in Eq. 5; the full GRPO objective is provided in Appendix C.

Interleaved Thinking. R1-like methods employ an iterative process of reasoning and retrieval to synthesize a final output. This process is modeled as an action sequence $\mathcal{A} = [a_0, a_1, \dots, a_t]$, initialized with $a_0 = \textbf{Think}$. For subsequent steps $i > 0$, the transition logic dictates that if the preceding action a_{i-1} was a reasoning step (**Think**), the subsequent action a_i must be either **Search** or **Answer**. Conversely, if $a_{i-1} \neq \textbf{Think}$, the system defaults to **Think**. The action space is defined as follows:

- **Think:** Derives reasoning steps utilizing existing internal knowledge.
- **Search:** Queries an external knowledge base to retrieve supplementary information.
- **Answer:** Terminates the sequence by providing the final response once information sufficiency is achieved.
- **No-Think:** Introduced in AdaptR1, this operator permits the model to bypass the explicit reasoning phase during a scheduled **No-Think** step.

4 AdaptR1

AdaptR1 extends GRPO to train an interleaved QA policy that can decide at each scheduled reasoning slot whether to generate an explicit rationale or emit the **No-Think** token

<think>no_think</think>. This step-wise decision is important for multi-hop QA: a trajectory may need explicit reasoning after some retrieval results, but not after every intermediate step. We train this behavior directly with RL rather than SFT, since constructing oracle trajectories that label exactly when reasoning should be skipped is ambiguous and dataset-dependent.

4.1 Quality-Gated Efficiency Objective

AdaptR1 is designed to optimize efficiency under an answer-quality constraint rather than as a pure length penalty. Let $R_{\text{ans}}(o) \in [0, 1]$ denote the answer reward of a generated trajectory o , measured by F1, and let $R_{\text{nt}}(o) \in [0, 1]$ denote a bounded efficiency reward derived from the number of No-Think actions. The intended objective is to reward efficiency only inside the feasible region of sufficiently accurate answers:

$$\max_{\pi_{\theta}} \mathbb{E}_{o \sim \pi_{\theta}} [R_{\text{nt}}(o)] \quad \text{s.t.} \quad R_{\text{ans}}(o) \geq \tau. \quad (1)$$

Operationally, we implement this constraint with a threshold mask and scale the efficiency bonus by the answer reward:

$$r_{\text{AdaptR1}}(o) = R_{\text{ans}}(o) (1 + \omega R_{\text{nt}}(o) \cdot \mathbb{I}[R_{\text{ans}}(o) \geq \tau]). \quad (2)$$

Eq. 2 is a compact view of the objective: the threshold mask prevents low-quality trajectories from receiving positive efficiency gradients, while the clipping and KL penalty in GRPO retain the trust-region-style stabilization of the base optimizer.

4.2 AdaptR1 Reward

As illustrated in Figure 2, our default implementation uses an absolute No-Think reward. Let $n_{\text{nt}}(o)$ be the number of No-Think actions in trajectory o . We first compute an uncapped efficiency bonus and then bound it by 1:

$$r_{\text{nt}}(o) = n_{\text{nt}}(o) \times r_0, \quad (3)$$

$$R_{\text{nt}}(o) = \min(r_{\text{nt}}(o), 1), \quad (4)$$

where r_0 is the unit reward for skipping one reasoning round. The actual training reward is the implementation form of Eq. 2:

$$r_{\text{AdaptR1}}(o) = r_{\text{answer}}(o) + \mathbb{I}[r_{\text{answer}}(o) \geq \tau] \cdot \omega r_{\text{answer}}(o) R_{\text{nt}}(o). \quad (5)$$

Here, τ is the answer-quality gate and ω controls the strength of the efficiency bonus relative to the

answer reward. The ceiling on R_{nt} keeps the No-Think signal auxiliary, so the model is rewarded for concise trajectories only when answer quality remains acceptable. The hyperparameters τ and ω are evaluated in the ablation studies.

4.3 Step-Wise Weighting and Reward Variants

The main experiments use the absolute No-Think reward in Eq. 5. To make this reward position-aware, we replace $R_{\text{nt}}(o)$ with a step-wise weighted efficiency reward. Let $n_{\text{nt}}^{(j)}(o)$ count No-Think actions at step j . Since the average trajectory length in our datasets is typically two to three rounds, we separate the first step from later steps:

$$s_{\lambda}(o) = \lambda n_{\text{nt}}^{(1)}(o) + (1 - \lambda) \sum_{j>1} n_{\text{nt}}^{(j)}(o), \quad (6)$$

$$R_{\text{nt}}^{\lambda}(o) = \min(r_0 s_{\lambda}(o), 1).$$

The coefficient λ controls whether the reward pressure favors early or later No-Think actions. We set $\lambda = 0.9$ in the main experiments and study its sensitivity in Section 6.

For reward-shape ablations, we additionally compare against a relative variant that normalizes the efficiency bonus by the total number of rounds:

$$p_{\text{nt}}(o) = \frac{n_{\text{nt}}(o)}{n_{\text{all}}(o)}. \quad (7)$$

This variant uses the same answer-quality gate as AdaptR1:

$$r_{\text{rel}}(o) = r_{\text{answer}}(o) + \mathbb{I}[r_{\text{answer}}(o) \geq \tau] \cdot \omega r_{\text{answer}}(o) p_{\text{nt}}(o). \quad (8)$$

This relative reward is not the default AdaptR1 objective; it is included to test whether ratio-based normalization encourages more stable exploration than the absolute bounded bonus.

5 Experiments

5.1 Setups

Datasets and Metrics. Following Graph-R1 (Luo et al., 2025a), we conduct experiments on six common QA datasets (Jin et al., 2025b): 2Wiki-hop (Ho et al., 2020), HotpotQA (Yang et al., 2018), Musique (Trivedi et al., 2022b), NQ (Kwiatkowski et al., 2019), PopQA (Mallen et al., 2023), and TriviaQA (Joshi et al., 2017). To keep comparisons controlled, all training-based methods use the same

Method	2Wiki.		HotpotQA		Musique		NQ		PopQA		TriviaQA		Avg.		
	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	R-S
<i>GPT-4o-mini</i>															
⊗ NaiveGeneration	4.69	17.03	18.75	31.79	3.13	11.45	2.34	21.59	10.36	25.95	28.91	47.73	11.36	25.92	-
▣ StandardRAG	7.03	22.31	35.16	46.70	9.38	17.31	7.03	26.85	18.75	30.58	31.25	48.55	18.10	32.05	52.68
⊗ GraphRAG	3.91	16.02	19.53	31.67	7.03	15.14	3.91	20.31	8.59	20.92	32.03	45.13	12.50	24.87	32.48
⊗ LightRAG	3.13	16.59	18.75	30.70	3.91	14.39	2.34	19.09	5.47	24.47	25.00	40.18	9.77	24.24	47.42
⊗ PathRAG	3.91	12.42	10.94	23.12	3.13	11.49	2.34	20.01	2.34	15.65	19.53	37.44	7.03	20.02	46.71
⊗ HippoRAG2	7.03	16.27	19.53	31.78	6.25	12.37	7.81	24.56	9.38	21.10	32.81	48.86	13.80	25.82	36.41
⊗ HyperGraphRAG	4.69	21.14	21.88	37.46	6.25	20.40	3.91	22.95	13.28	29.48	28.91	44.95	13.15	29.40	61.82
<i>Qwen2.5-7B-Instruct</i>															
⊗ NaiveGeneration	3.12	12.25	6.25	18.58	0.00	4.06	1.56	13.00	0.78	12.82	7.03	24.51	3.12	14.20	-
▣ StandardRAG	7.81	12.75	10.16	21.10	0.78	4.53	1.56	15.97	3.12	13.10	8.59	24.90	5.34	15.39	52.67
⊗ SFT	11.72	20.28	19.53	27.59	5.47	10.02	5.12	19.02	20.31	27.93	31.25	39.21	15.57	24.01	-
⊗ R1	25.00	30.99	31.25	37.05	7.03	14.53	16.41	28.45	26.56	30.35	49.22	57.33	25.91	33.12	-
▣ R1-Searcher	27.34	33.96	39.84	46.36	10.16	16.63	32.03	44.93	41.41	47.12	56.25	64.76	34.51	42.29	51.26
▣ Search-R1	35.15	38.21	43.77	51.26	17.18	21.45	38.34	43.79	43.75	47.03	51.56	61.03	38.29	43.80	53.06
▣ Search-AdaptR1	45.31	51.55	47.66	53.68	25.00	34.31	33.59	47.13	42.19	46.80	61.72	70.63	42.58	50.68	65.13
⊗ Graph-R1	58.59	68.18	55.47	63.55	37.50	48.33	35.16	49.55	50.78	54.01	66.41	72.02	50.65	59.27	60.46
⊗ Graph-AdaptR1	61.72	69.20	57.81	64.39	39.84	53.42	35.94	49.62	49.22	55.03	64.06	72.77	51.43	60.74	61.88

Table 1: Main results under the controlled multi-hop QA setting with best in **bold**. ⊗ means prompt engineering, ▣ means training, ⊗ means no knowledge interaction, ▣ means chunk-based knowledge, and ⊗ means graph-based knowledge.

fixed split of 5,120 training and 128 testing instances per dataset. We use EM and F1 to evaluate answer quality, and R-S to evaluate retrieval performance.

Baselines. We consider both training-free and training-based baselines. The training-free methods include NaiveGeneration, StandardRAG (Lewis et al., 2020), GraphRAG (Edge et al., 2025), LightRAG (Guo et al., 2025), PathRAG (Chen et al., 2025a), HippoRAG2 (Gutiérrez et al., 2025), and HyperGraphRAG (Luo et al., 2025b). The training-based baselines include SFT (Zheng et al., 2024), R1 (Shao et al., 2024), R1-Searcher (Song et al., 2025), Search-R1 (Jin et al., 2025a), and Graph-R1 (Luo et al., 2025a). For the key RL comparisons, Search-R1, Graph-R1, Search-AdaptR1, and Graph-AdaptR1 are initialized from the same Qwen2.5-7B-Instruct (Qwen et al., 2025) backbone and trained under the same data split and hyperparameter budget; AdaptR1 is not initialized from trained Search-R1 or Graph-R1 checkpoints. This isolates the effect of the adaptive reward from differences in data, initialization, and training overhead.

Implementation Details. We instantiate AdaptR1 on two baselines, Search-R1 and Graph-R1, yielding Search-AdaptR1 and Graph-AdaptR1. Unless otherwise stated, we use the absolute No-

Think reward with weight $\lambda = 0.9$, threshold $\tau = 0.6$ and coefficient $\omega = 0.2$. Detailed implementations of baselines and our method are provided in Appendix F.

Retriever. The retriever follows the corresponding backbone method. Search-R1 uses E5 (Wang et al., 2022), while Graph-R1 uses hypergraph-based retrieval with bge-large-en-v1.5 (Chen et al., 2023).

5.2 Main Experiments

We observe that Graph-AdaptR1 yields results comparable to Graph-R1, achieving an average F1 improvement of 1.5. Search-AdaptR1 outperforms Search-R1 by a larger margin, with an average F1 increase of 6.9. This consistent improvement across datasets and retrieval pipelines suggests that AdaptR1 is not tied to a single dataset or retrieval design. The improved R-S scores further indicate that adaptive skipping can preserve, and in some cases improve, the retrieval behavior needed for accurate answers.

Think Token Economy. To evaluate the think token economy of AdaptR1, we analyze the token consumption detailed in Table 2. Under the Graph-R1 setting, Graph-AdaptR1 lowers average think tokens from 102.07 to 30.92, corresponding to a 69.71% reduction, while maintaining or slightly improving average F1. The largest reduction ap-

Model	2Wiki	HotpotQA	Musique	NQ	PopQA	TriviaQA	Avg.
Search-R1	155.00	100.74	117.23	95.56	105.78	92.63	111.16
Search-AdaptR1	119.21(↓23.09%)	64.73(↓35.75%)	82.52(↓29.61%)	60.27(↓36.93%)	42.60(↓59.73%)	90.77(↓2.01%)	76.68(↓31.02%)
Graph-R1	151.14	103.68	148.29	61.99	54.01	93.33	102.07
Graph-AdaptR1	58.92(↓61.02%)	10.00(↓90.35%)	86.06(↓41.97%)	10.00(↓83.87%)	10.00(↓81.48%)	10.52(↓88.73%)	30.92(↓69.71%)

Table 2: Comparisons of **Think Tokens** before and after applying AdaptR1 in Search-R1 and Graph-R1 settings.

pears on HotpotQA, where think tokens decrease from 103.68 to 10.00, a 90.35% reduction. This drastic decrease in token usage does not come at the cost of accuracy, as shown in the main experiments. Instead, it highlights that Graph-R1 contains substantial redundant explicit reasoning steps. By eliminating these superfluous reasoning steps, AdaptR1 lowers computational cost and latency while maintaining or improving performance.

5.3 Case Study

To provide a granular understanding of the model’s behavior, we present a comparison of generation trajectories with and without AdaptR1 in Section E.4. The case study illustrates that standard LLMs can exhibit “over-thinking,” generating exhaustive and sometimes circular reasoning chains even in multi-hop QA. In contrast, AdaptR1 bypasses redundant thinking while preserving the key reasoning link needed for the answer. We additionally analyze cases where skipping thinking hurts answer quality in Section E.5. These qualitative results corroborate our quantitative efficiency results while making clear that no_think is beneficial when used selectively rather than as an unconditional rule.

6 Ablations and Analysis

In this section, we validate the main design choices of **AdaptR1** through a prompt-only control experiment, a temporal distribution analysis, and reward-design ablations. All analyses use Graph-R1 as the backbone.

6.1 Necessity of RL Training

To verify that adaptive skipping is learned through RL rather than triggered by the prompt alone, we compare the base model before training with the RL-trained Graph-AdaptR1 model under the same adaptive prompt. As shown in Table 3, the base model rarely uses no_think on complex datasets and performs poorly. RL training substantially improves both answer quality and adaptive skipping; for example, on Musique, the No-Think rate increases from 13.02% to 50.67%, while F1 improves

from 8.41 to 53.42. This indicates that no_think becomes useful only after the model learns how to integrate it into the reason-search-answer trajectory.

6.2 Step-wise Adaptive Thinking and Temporal Distribution

We first evaluate the temporal dynamics of the No-Think mechanism to understand how the model allocates its reasoning budget across different stages of the problem-solving trajectory. Table 4 presents the average step-wise No-Think ratio with a balanced penalty factor ($\lambda = 0.5$) for Musique. Other datasets are shown in E.2. Compared with saturated datasets such as HotpotQA, NQ, PopQA, and TriviaQA, Musique exposes more complex reasoning dynamics and is therefore the most diagnostic setting for studying where explicit thinking remains necessary.

Balanced Strategy: The results reveal a distinct, emergent behavior: the model predominantly learns to bypass extensive reasoning during the initial steps (Steps 1–2), reserving its computational budget for the final stages of the trajectory (Steps 3–4). This observation is somewhat counter-intuitive; one might expect the initial planning phase to require significant cognitive load. However, the data suggests that for multi-hop QA tasks, the model adopts a “retrieve-then-reason” strategy. The early steps likely involve schema activation or direct information retrieval that can be handled heuristically, whereas the final steps require synthesis and deduction to formulate the answer. We therefore examine whether larger early-step weights better match this behavior.

Sensitivity to λ : We further explore the impact of varying the step-wise reward weight λ from 0.1 to 0.9.

- *Low λ (0.1–0.4):* Assigning insufficient reward to early No-Think actions generally leads to lower F1, suggesting that the model may still over-reason on simple intermediate steps.
- *High λ (0.6–0.9):* Larger λ often improves

Metric	Model Status	2Wiki	HotpotQA	Musique	NQ	PopQA	TriviaQA
No-Think Rate (%)	Before Training	13.05	11.22	13.02	23.28	31.11	20.41
	After Training	61.74	100.00	50.67	100.00	100.00	99.61
F1 Score	Before Training	13.51	19.15	8.41	13.46	17.18	26.67
	After Training	69.20	64.39	53.42	49.62	55.03	72.77

Table 3: Comparison before and after RL training using the same adaptive prompt.

	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Avg. Ratio	F1
<i>Impact of Step-wise Weight (λ)</i>								
$\lambda = 0.5$	1.0000	0.9688	0.0391	0.0444	0.0000	0.0000	0.5294	0.4923
<i>Sensitivity Analysis</i>								
$\lambda = 0.1$	0.0000	0.0000	0.0000	0.0000	0.0000	-	0.0000	0.5135
$\lambda = 0.2$	0.0000	0.0000	0.0156	0.0000	0.0000	0.0000	0.0047	0.4925
$\lambda = 0.3$	1.0000	0.0000	0.9297	0.4955	0.5000	0.0000	0.6043	0.4923
$\lambda = 0.4$	0.0000	0.0709	0.9762	0.9043	0.6667	1.0000	0.4780	0.4814
$\lambda = 0.6$	1.0000	0.1797	0.0106	0.0000	0.0000	-	0.4053	0.4724
$\lambda = 0.7$	0.0000	0.0000	0.0000	0.0000	-	-	0.0000	0.5235
$\lambda = 0.8$	1.0000	1.0000	0.9453	0.9500	0.8696	1.0000	0.9705	0.4743
$\lambda = 0.9$	1.0000	0.0000	0.8359	0.1983	0.3333	0.0000	0.5067	0.5342

Table 4: Analysis of the No-Think ratio from step 1 to 6 and performance (F1) for Graph-R1 with and without AdaptR1 on Musique. The parameter λ controls the step-wise penalty weight. We observe the temporal distribution of token savings across sequential reasoning steps.

over low λ , and $\lambda = 0.9$ gives the best F1 on Musique. The model consistently skips the first step but still uses explicit thinking later when retrieved evidence must be synthesized, supporting a retrieve-then-reason strategy rather than indiscriminate skipping.

These findings support the step-wise reward design, so we use $\lambda = 0.9$ in the main experiments in Table 1.

6.3 Design of the Adaptive Reward Function

The efficacy of Reinforcement Learning (RL) is heavily contingent on reward shaping. In this subsection, we investigate the specific design components of the AdaptR1 reward structure, including formulation (absolute vs. relative), constraints (ceilings), and hyperparameter sensitivity.

	2Wiki	HotpotQA	Musique	NQ	PopQA	TriviaQA	Avg.
AdaptR1	69.20	64.39	53.42	49.62	55.03	72.77	60.74
AdaptR1-rel	64.32	66.19	45.12	48.94	53.46	71.44	58.25
w/o Top Ceiling	63.29	61.86	46.37	47.38	49.19	69.12	56.20

Table 5: F1 scores of Graph-AdaptR1 variants across varying datasets. We contrast the standard formulation against relative rewards and no ceiling rewards.

Absolute vs. Relative Reward Formulation We compare our absolute No-Think reward against a

relative formulation (Table 5). A relative reward scales with the fraction of skipped steps, which appears smoother but is easier to exploit: the model can shorten the trajectory and skip a larger ratio of steps regardless of context. In contrast, the absolute formulation preserves the multi-turn reasoning structure and yields better average F1.

Impact of Reward Ceiling We further test the necessity of a “top ceiling,” a hard limit on the accumulation of efficiency rewards. As shown in the “w/o top ceiling” row of Table 5, removing this constraint degrades performance significantly. Without a ceiling, the efficiency reward can dominate the optimization landscape and encourage reward hacking. The ceiling keeps efficiency as a secondary objective that should not override answer accuracy.

Threshold Sensitivity (τ) The threshold τ determines the confidence level required for the model to trigger a No-Think action. Table 6 illustrates the trade-off:

- *Low τ (< 0.6):* A permissive threshold results in frequent, unjustified skipping of reasoning steps, harming performance (Avg F1 ≈ 55.3).
- *High τ (> 0.8):* An overly strict threshold renders the No-Think reward too sparse. The

model rarely attempts to skip, negating the efficiency benefits of AdaptR1.

Our results identify $\tau = 0.6$ as the critical inflection point where the model reliably filters unnecessary reasoning without truncating valid cognitive processes.

τ	2Wiki	HotpotQA	Musique	NQ	PopQA	TriviaQA	Avg.
0.2	65.79	62.29	48.77	49.12	52.45	72.14	58.43
0.4	65.22	65.38	47.69	49.90	55.89	70.14	59.04
0.6	69.20	64.39	53.42	49.62	55.03	72.77	60.74
0.8	64.79	64.53	44.14	50.58	56.15	70.20	58.40
1.0	67.37	67.13	46.40	48.99	55.29	70.52	59.28

Table 6: Ablation study on the confidence threshold τ with the ratio fixed at 0.2. Performance peaks at $\tau = 0.6$, suggesting a balance between aggressive skipping and conservative reasoning.

Reward Coefficient (ω) Finally, we analyze the magnitude of the efficiency reward relative to the correctness reward, controlled by coefficient ω (Table 7). The data exhibits an inverted U-shaped curve. A small ω (0.1) provides a weak adaptive signal, while a large ω (> 0.4) distracts optimization from the primary QA objective. We find that $\omega = 0.2$ provides the best trade-off, suggesting that the efficiency signal should remain auxiliary to answer correctness.

ω	2Wiki	HotpotQA	Musique	NQ	PopQA	TriviaQA	Avg.
0.1	63.75	65.30	48.11	47.68	49.09	71.13	57.51
0.2	69.20	64.39	53.42	49.62	55.03	72.77	60.74
0.3	62.56	60.98	45.95	48.29	47.68	69.44	55.82
0.4	65.72	63.71	45.21	51.95	54.22	70.82	58.61
0.5	64.80	67.27	46.63	47.42	51.18	70.47	57.96
0.6	62.24	58.39	49.54	48.87	47.76	68.60	55.90

Table 7: Ablation study on the reward coefficient ω . The optimal value $\omega = 0.2$ indicates that efficiency rewards must be carefully scaled relative to answer accuracy rewards.

Taken together, the ablations provide practical guardrails for avoiding reward hacking: τ should not be lower than 0.6, ω is most reliable around 0.1–0.2, and λ is best treated as a step-wise pressure term rather than a universal instruction to skip. These ranges preserve the answer reward as the dominant objective while allowing the model to discover efficient trajectories.

6.4 Training Dynamics

To better understand the learning progression of AdaptR1, we visualize No-Think behavior on the difficult Musique dataset in Figure 3. The curve

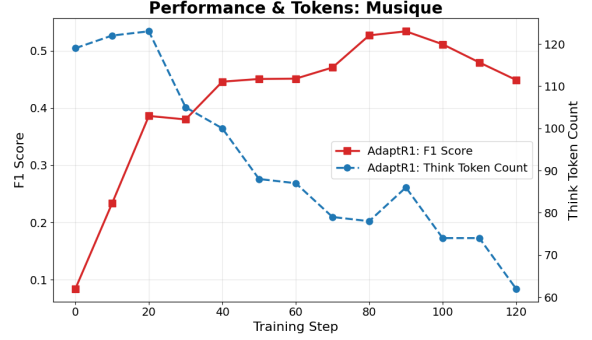


Figure 3: Training dynamics on Musique. The evolution of No-Think behavior indicates a phased learning process.

shows a phased rather than monotonic trade-off between accuracy and reasoning length. In early training (steps 0–20), F_1 rises rapidly to about 0.4 while think tokens remain high (~ 120), suggesting that the model first learns to solve the task with ample reasoning. During steps 20–80, No-Think behavior emerges: F_1 peaks above 0.6 as think tokens decline toward ~ 80 , showing that AdaptR1 learns to prune redundant thinking after acquiring task competence. Past step 90, however, tokens fall further to ~ 60 while F_1 drops to about 0.45, indicating that excessive pruning can skip necessary synthesis. Appendix E.3 shows the same trend on other datasets; most reduce think tokens early and then improve accuracy, while 2Wiki briefly recovers more thinking near step 80.

7 Conclusion

In this paper, we presented **AdaptR1**, an RL-based framework for mitigating “over-thinking” in multi-hop question answering. Unlike adaptive methods that rely on global routing decisions or SFT cold-start trajectories, AdaptR1 learns a fine-grained, step-wise policy that decides when to reason explicitly, when to query external knowledge, and when to skip redundant thinking at each intermediate stage. Empirically, AdaptR1 improves both Search-R1 and Graph-R1 settings: Search-AdaptR1 raises average F1 from 43.80 to 50.68, while Graph-AdaptR1 raises average F1 from 59.27 to 60.74. Under the Graph-R1 setting, it reduces average think tokens by 69.71%, with the largest per-dataset reduction reaching 90.35%, while maintaining or slightly improving answer performance. Our analyses further show that adaptive skipping is learned through RL, that overthinking is concentrated in the initial planning stages rather than the

final synthesis stage, and that quality-gated rewards are important for avoiding reward hacking. These results suggest that RL-only adaptive interleaved thinking is a promising direction for efficient multi-hop reasoning.

Limitations

The limitations of AdaptR1 primarily stem from its sensitivity to hyperparameters and potential training instability, as the method relies on balancing the confidence threshold (τ), reward coefficient (ω), and step-wise weights (λ). Our ablations identify useful ranges, but overly aggressive settings can still cause the model to over-prune essential reasoning steps and prioritize brevity over correctness in later training epochs. In addition, our current scope is multi-hop QA, where trajectories are typically short to medium length. We do not claim that the same reward design directly transfers to DeepResearch-style tasks that require much longer planning and sustained reasoning; extending AdaptR1 to that setting remains future work.

References

- Pranjal Aggarwal and Sean Welleck. 2025. L1: Controlling how long a reasoning model thinks with reinforcement learning. *arXiv preprint arXiv:2503.04697*.
- Daman Arora and Andrea Zanette. 2025. [Training language models to reason efficiently](#). *CoRR*, abs/2502.04463.
- Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2024. Self-rag: Learning to retrieve, generate, and critique through self-reflection.
- Boyuan Chen, Zirui Guo, Zidan Yang, Yuluo Chen, Junze Chen, Zhenghao Liu, Chuan Shi, and Cheng Yang. 2025a. [Pathrag: Pruning graph-based retrieval augmented generation with relational paths](#). *Preprint*, arXiv:2502.14902.
- Jianlv Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. 2023. [Bge m3-embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation](#). *Preprint*, arXiv:2309.07597.
- Qianben Chen, Jingyi Cao, Jiayu Zhang, Tianrui Qin, Xiaowan Li, King Zhu, Dingfeng Shi, He Zhu, Minghao Liu, Xiaobo Liang, Xin Gui, Ge Zhang, Jian Yang, Yuchen Eleanor Jiang, and Wangchunshu Zhou. 2025b. [A²fm: An adaptive agent foundation model for tool-aware hybrid reasoning](#). *Preprint*, arXiv:2510.12838.
- Xingyu Chen, Jiahao Xu, Tian Liang, Zhiwei He, Jianhui Pang, Dian Yu, Linfeng Song, Qiuzhi Liu, Mengfei Zhou, Zhuosheng Zhang, Rui Wang, Zhaopeng Tu, Haitao Mi, and Dong Yu. 2024. [Do NOT think that much for 2+3=? on the overthinking of o1-like llms](#). *CoRR*, abs/2412.21187.
- Ken Deng, Zizheng Zhan, Wen Xiang, Wenqiang Zhu, Weihao Li, Jingxuan Xu, Tianhao Peng, Xinpeng Lei, Kun Wu, Yifan Yao, Haoyang Huang, Huaixi Tang, Kepeng Lei, Zhiyi Lai, Songwei Yu, Zongxian Feng, Zuchen Gao, Weihao Xie, Chenchen Zhang, and 9 others. 2025. [Hipo: Hybrid policy optimization for dynamic reasoning in llms](#). *Preprint*, arXiv:2509.23967.
- Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, Dasha Metropolitan, Robert Osazuwa Ness, and Jonathan Larson. 2025. [From local to global: A graph rag approach to query-focused summarization](#). *Preprint*, arXiv:2404.16130.
- Xinyan Guan, Jiali Zeng, Fandong Meng, Chunlei Xin, Yaojie Lu, Hongyu Lin, Xianpei Han, Le Sun, and Jie Zhou. 2025. Deeprag: Thinking to retrieve step by step for large language models. *arXiv preprint arXiv:2502.01142*.
- Zirui Guo, Lianghao Xia, Yanhua Yu, Tu Ao, and Chao Huang. 2025. [Lightrag: Simple and fast retrieval-augmented generation](#). *Preprint*, arXiv:2410.05779.
- Bernal Jiménez Gutiérrez, Yiheng Shu, Weijian Qi, Sizhe Zhou, and Yu Su. 2025. [From rag to memory: Non-parametric continual learning for large language models](#). *Preprint*, arXiv:2502.14802.
- Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. 2020. [Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps](#). In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 6609–6625, Barcelona, Spain (Online). International Committee on Computational Linguistics.
- Bairu Hou, Yang Zhang, Jiabao Ji, Yujian Liu, Kaizhi Qian, Jacob Andreas, and Shiyu Chang. 2025. [Thinkprune: Pruning long chain-of-thought of llms via reinforcement learning](#). *Preprint*, arXiv:2504.01296.
- Zhengbao Jiang, Frank F Xu, Luyu Gao, Zhiqing Sun, Qian Liu, Jane Dwivedi-Yu, Yiming Yang, Jamie Callan, and Graham Neubig. 2023. Active retrieval augmented generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 7969–7992.
- Bowen Jin, Hansi Zeng, Zhenrui Yue, Jinsung Yoon, Sercan Arik, Dong Wang, Hamed Zamani, and Jiawei Han. 2025a. [Search-r1: Training llms to reason and leverage search engines with reinforcement learning](#). *Preprint*, arXiv:2503.09516.

- Jiajie Jin, Yutao Zhu, Zhicheng Dou, Guanting Dong, Xinyu Yang, Chenghao Zhang, Tong Zhao, Zhao Yang, and Ji-Rong Wen. 2025b. [Flashrag: A modular toolkit for efficient retrieval-augmented generation research](#). In *Companion Proceedings of the ACM on Web Conference 2025*, WWW '25, page 737–740, New York, NY, USA. Association for Computing Machinery.
- Mandar Joshi, Eunsol Choi, Daniel Weld, and Luke Zettlemoyer. 2017. [TriviaQA: A large scale distantly supervised challenge dataset for reading comprehension](#). In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1601–1611, Vancouver, Canada. Association for Computational Linguistics.
- Yu Kang, Xianghui Sun, Liangyu Chen, and Wei Zou. 2025. C3ot: Generating shorter chain-of-thought without compromising effectiveness. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 24312–24320.
- Abhinav Kumar, Jaechul Roh, Ali Naseh, Marzena Karpinska, Mohit Iyyer, Amir Houmansadr, and Eugene Bagdasarian. 2025. Overthink: Slow-down attacks on reasoning llms. *arXiv preprint arXiv:2502.02542*.
- Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, Kristina Toutanova, Llion Jones, Matthew Kelcey, Ming-Wei Chang, Andrew M. Dai, Jakob Uszkoreit, Quoc Le, and Slav Petrov. 2019. [Natural questions: A benchmark for question answering research](#). *Transactions of the Association for Computational Linguistics*, 7:452–466.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. [Retrieval-augmented generation for knowledge-intensive nlp tasks](#). In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474. Curran Associates, Inc.
- Yuan Li, Qi Luo, Xiaonan Li, Bufan Li, Qinyuan Cheng, Bo Wang, Yining Zheng, Yuxin Wang, Zhangyue Yin, and Xipeng Qiu. 2025. R3-rag: Learning step-by-step reasoning and retrieval for llms via reinforcement learning. *arXiv preprint arXiv:2505.23794*.
- Chenwei Lou, Zewei Sun, Xinnian Liang, Meng Qu, Wei Shen, Wenqi Wang, Yuntao Li, Qingping Yang, and Shuangzhi Wu. 2025. [Adacot: Pareto-optimal adaptive chain-of-thought triggering via reinforcement learning](#). *Preprint*, arXiv:2505.11896.
- Haoran Luo, Guanting Chen, Qika Lin, Yikai Guo, Fangzhi Xu, Zemin Kuang, Meina Song, Xiaobao Wu, Yifan Zhu, Luu Anh Tuan, and 1 others. 2025a. Graph-r1: Towards agentic graphrag framework via end-to-end reinforcement learning. *arXiv preprint arXiv:2507.21892*.
- Haoran Luo, Haihong E, Guanting Chen, Yandan Zheng, Xiaobao Wu, Yikai Guo, Qika Lin, Yu Feng, Zemin Kuang, Meina Song, Yifan Zhu, and Luu Anh Tuan. 2025b. [Hypergraphrag: Retrieval-augmented generation via hypergraph-structured knowledge representation](#). *Preprint*, arXiv:2503.21322.
- Haotian Luo, Li Shen, Haiying He, Yibo Wang, Shiwei Liu, Wei Li, Naiqiang Tan, Xiaochun Cao, and Dacheng Tao. 2025c. [O1-pruner: Length-harmonizing fine-tuning for o1-like reasoning pruning](#). *CoRR*, abs/2501.12570.
- Wenjie Ma, Jingxuan He, Charlie Snell, Tyler Griggs, Sewon Min, and Matei Zaharia. 2025a. Reasoning models can be effective without thinking. *arXiv preprint arXiv:2504.09858*.
- Xinyin Ma, Guangnian Wan, Runpeng Yu, Gongfan Fang, and Xinchao Wang. 2025b. [Cot-valve: Length-compressible chain-of-thought tuning](#). *Preprint*, arXiv:2502.09601.
- Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. [When not to trust language models: Investigating effectiveness of parametric and non-parametric memories](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9802–9822, Toronto, Canada. Association for Computational Linguistics.
- Reiichiro Nakano, Jacob Hilton, Suchir Balaji, Jeff Wu, Long Ouyang, Christina Kim, Christopher Hesse, Shantanu Jain, Vineet Kosaraju, William Saunders, and 1 others. 2021. Webgpt: Browser-assisted question-answering with human feedback. *arXiv preprint arXiv:2112.09332*.
- Sania Nayab, Giulio Rossolini, Marco Simoni, Andrea Saracino, Giorgio Buttazzo, Nicolamaria Manes, and Fabrizio Giacomelli. 2024. Concise thoughts: Impact of output length on llm reasoning and cost. *arXiv preprint arXiv:2407.19825*.
- OpenAI, :, Aaron Hurst, Adam Lerer, Adam P. Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, Aleksander Madry, Alex Baker-Whitcomb, Alex Beutel, Alex Borzunov, Alex Carney, Alex Chow, Alex Kirillov, and 21 others. 2024a. [Gpt-4o system card](#). *Preprint*, arXiv:2410.21276.
- OpenAI, :, Aaron Jaech, Adam Kalai, Adam Lerer, Adam Richardson, Ahmed El-Kishky, Aiden Low, Alec Helyar, Aleksander Madry, Alex Beutel, Alex Carney, Alex Ifimie, Alex Karpenko, Alex Tachard Passos, Alexander Neitz, Alexander Prokofiev, Alexander Wei, Allison Tam, and 19 others. 2024b. [Openai o1 system card](#). *Preprint*, arXiv:2412.16720.
- Ofir Press, Muru Zhang, Sewon Min, Ludwig Schmidt, Noah A. Smith, and Mike Lewis. 2023. [Measuring and narrowing the compositionality gap in language models](#). *Preprint*, arXiv:2210.03350.

- Qwen, :, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiayi Yang, Jingren Zhou, and 25 others. 2025. [Qwen2.5 technical report](#). *Preprint*, arXiv:2412.15115.
- Zhihong Shao, Yeyun Gong, Yelong Shen, Minlie Huang, Nan Duan, and Weizhu Chen. 2023. [Enhancing retrieval-augmented large language models with iterative retrieval-generation synergy](#). *Preprint*, arXiv:2305.15294.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. 2024. [Deepseekmath: Pushing the limits of mathematical reasoning in open language models](#). *Preprint*, arXiv:2402.03300.
- Yi Shen, Jian Zhang, Jieyun Huang, Shuming Shi, Wenjing Zhang, Jiangze Yan, Ning Wang, Kai Wang, and Shiguo Lian. 2025. [DAST: difficulty-adaptive slow-thinking for large reasoning models](#). *CoRR*, abs/2503.04472.
- Guangming Sheng, Chi Zhang, Zilingfeng Ye, Xibin Wu, Wang Zhang, Ru Zhang, Yanghua Peng, Haibin Lin, and Chuan Wu. 2025. [Hybridflow: A flexible and efficient rlhf framework](#). In *Proceedings of the Twentieth European Conference on Computer Systems*, EuroSys '25, page 1279–1297. ACM.
- Huatong Song, Jinhao Jiang, Yingqian Min, Jie Chen, Zhipeng Chen, Wayne Xin Zhao, Lei Fang, and Jirong Wen. 2025. [R1-searcher: Incentivizing the search capability in llms via reinforcement learning](#). *Preprint*, arXiv:2503.05592.
- Yang Sui, Yu-Neng Chuang, Guanchu Wang, Jiamu Zhang, Tianyi Zhang, Jiayi Yuan, Hongyi Liu, Andrew Wen, Shaochen Zhong, Na Zou, and 1 others. 2025. [Stop overthinking: A survey on efficient reasoning for large language models](#). *arXiv preprint arXiv:2503.16419*.
- Kimi Team, Angang Du, Bofei Gao, Bowei Xing, Changjiu Jiang, Cheng Chen, Cheng Li, Chenjun Xiao, Chenzhuang Du, Chonghua Liao, Chunling Tang, Congcong Wang, Dehao Zhang, Enming Yuan, Enzhe Lu, Fengxiang Tang, Flood Sung, Guangda Wei, Guokun Lai, and 75 others. 2025. [Kimi k1.5: Scaling reinforcement learning with llms](#). *CoRR*, abs/2501.12599.
- Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2022a. [Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions](#). *arXiv preprint arXiv:2212.10509*.
- Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2022b. [MuSiQue: Multi-hop questions via single-hop question composition](#). *Transactions of the Association for Computational Linguistics*, 10:539–554.
- Songjun Tu, Jiahao Lin, Qichao Zhang, Xiangyu Tian, Linjing Li, Xiangyuan Lan, and Dongbin Zhao. 2025. [Learning when to think: Shaping adaptive reasoning in rl-style models via multi-stage rl](#). *Preprint*, arXiv:2505.10832.
- Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, and Furu Wei. 2022. [Text embeddings by weakly-supervised contrastive pre-training](#). *arXiv preprint arXiv:2212.03533*.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, and 1 others. 2022. [Chain-of-thought prompting elicits its reasoning in large language models](#). *Advances in neural information processing systems*, 35:24824–24837.
- Han Wu, Yuxuan Yao, Shuqi Liu, Zehua Liu, Xiaojin Fu, Xiongwei Han, Xing Li, Hui-Ling Zhen, Tao Zhong, and Mingxuan Yuan. 2025a. [Unlocking efficient long-to-short LLM reasoning with model merging](#). *CoRR*, abs/2503.20641.
- Siye Wu, Jian Xie, Yikai Zhang, Aili Chen, Kai Zhang, Yu Su, and Yanghua Xiao. 2025b. [Arm: Adaptive reasoning model](#). *arXiv preprint arXiv:2505.20258*.
- Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. 2018. [HotpotQA: A dataset for diverse, explainable multi-hop question answering](#). In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380, Brussels, Belgium. Association for Computational Linguistics.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. [React: Synergizing reasoning and acting in language models](#). In *International Conference on Learning Representations (ICLR)*.
- Jiajie Zhang, Nianyi Lin, Lei Hou, Ling Feng, and Juanzi Li. 2025. [Adaptthink: Reasoning models can learn when to think](#). *Preprint*, arXiv:2505.13417.
- Yaowei Zheng, Richong Zhang, Junhao Zhang, Yanhan Ye, and Zheyang Luo. 2024. [LlamaFactory: Unified efficient fine-tuning of 100+ language models](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, pages 400–410, Bangkok, Thailand. Association for Computational Linguistics.
- Yuxiang Zheng, Dayuan Fu, Xiangkun Hu, Xiaojie Cai, Lyumanshan Ye, Pengrui Lu, and Pengfei Liu. 2025. [DeepResearcher: Scaling deep research via reinforcement learning in real-world environments](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 414–431, Suzhou, China. Association for Computational Linguistics.

A Prompt Template

Table 8 shows the Graph-AdaptR1 prompt. The template preserves the original reason-search-answer format while adding one adaptive instruction: when explicit reasoning is unnecessary, the model may emit <think>no_think</think> before either a query or final answer. Search-AdaptR1 uses the same adaptive instruction but keeps Search-R1’s native tool tags.

B Ethics Statement

This work utilizes publicly available datasets (2WikiMultiHopQA, HotpotQA, Musique, NQ, PopQA, and TriviaQA) that are widely used in the research community. We have adhered to the licenses and terms of use associated with these datasets. To the best of our knowledge, these datasets do not contain personally identifiable information (PII) or offensive content that would pose a risk to individuals. This study does not involve human subjects or human annotation, as all evaluations were conducted using automatic metrics.

A primary contribution of this work is the reduction of computational costs in Large Language Models (LLMs). By reducing the number of generated "think tokens" by up to 90% compared to standard reasoning methods, AdaptR1 significantly lowers the energy consumption and carbon footprint associated with model inference. This aligns with the goals of Green AI.

However, we acknowledge that our method relies on the pre-trained Qwen2.5-7B-Instruct model. Like all LLMs, this backbone model may carry inherent biases or the potential to generate toxic content derived from its training data. While our adaptive strategy aims to improve efficiency and does not explicitly introduce new biases, it does not actively mitigate existing ones. Users should exercise caution and implement appropriate safety guardrails when deploying such models in real-world applications.

AI assistants were used for language polishing, improving clarity and readability, and editorial integration of author-provided revision material. All scientific content, including the research ideas, methodology, experimental design, results, and conclusions, was conceived, implemented, and verified by the authors. The use of AI tools did not influence the technical decisions or the interpretation of experimental results.

C GRPO Objective

For each question $q \sim P(Q)$, GRPO samples a group of outputs $\{o_1, o_2, \dots, o_G\}$ from the old policy $\pi_{\theta_{\text{old}}}$ and updates the policy model π_{θ} by optimizing:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(O|q)} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left(\min(r_t(\theta) \hat{A}_{i,t}, \text{clip}(r_t(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_{i,t}) - \beta \mathbb{D}_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \right) \right]. \quad (9)$$

Here, $r_t(\theta) = \frac{\pi_{\theta}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}|q, o_{i,<t})}$ is the probability ratio, ε controls clipping, β regulates the KL penalty, and $\hat{A}_{i,t}$ is computed using group-relative reward:

$$\hat{A}_{i,t} = \frac{r_i - \text{mean}(\mathbf{r})}{\text{std}(\mathbf{r})}. \quad (10)$$

where $\mathbf{r} = \{r_1, r_2, \dots, r_G\}$ is the reward vector of G outputs. Since GRPO uses sequence-level rewards, $\hat{A}_{i,t}$ is constant across all tokens within the same trajectory.

D Algorithm

We show the algorithm for AdaptR1.

E Additional Experiments

E.1 Robustness Across Random Seeds

We further evaluate robustness with 5 random seeds on three representative datasets. Table 9 reports mean and standard deviation for answer metrics and think-token usage. Accuracy variance remains moderate, and the average think-token count remains substantially below the corresponding baseline in every dataset.

E.2 No-Think Ratios

We show the No-Think ratios of left 5 datasets in Table 10,11,12,13,14.

E.3 Training Dynamics

We show the training dynamics of six datasets in Figure 4.

E.4 Case Study

We show the case of Graph-R1 in Table 15 and Graph-AdaptR1 in Table 16.

Answer the given question. You can query from the knowledge base provided to you to answer the question. You can query knowledge as many times as you want. You can conduct reasoning inside `<think>...</think>` when needed. **If reasoning is not necessary, output `<think>no_think</think>` to skip reasoning.** If you need to query knowledge, set `{"query": <statement-to-search>}` between `<query>...</query>` after the `<think>...</think>` tags. When you have the final answer, output it inside `<answer>...</answer>` after the `<think>...</think>` tags. Please keep the answer short and clear. **Formats:** tool call with reasoning: `<think>...</think> <query>{"query": <statement-to-search>}</query>`; tool call without reasoning: `<think>no_think</think> <query>{"query": <statement-to-search>}</query>`; answer with reasoning: `<think>...</think> <answer>...</answer>`; answer without reasoning: `<think>no_think</think> <answer>...</answer>`.
Question: **question**. Assistant:

Table 8: Prompt template for Graph-AdaptR1. The red instruction introduces the adaptive No-Think action. Search-AdaptR1 uses the same adaptive instruction while preserving Search-R1’s native `<search>/<information>` interface.

Algorithm 1 AdaptR1

Require: Input x , LLM π_θ , Retrieval set $\mathcal{R}$, Max turns B .

Ensure: Output y .

```

1: Initialize  $y \leftarrow \emptyset$ 
2: Initialize  $b \leftarrow 0$ 
3: while  $b < B$  do
4:   Rollout  $y_b \leftarrow \emptyset$ 
5:   while True do
6:     Adaptively generate either a reasoning trace or <think>no_think</think>:  $y_t \sim \pi_\theta(\cdot \mid x, y + y_b)$ 
7:     concatenate token  $y_b \leftarrow y_b + y_t$ 
8:     if  $y_t$  in [</query>, </answer>, <eos>] then break
9:     end if
10:  end while
11:   $y \leftarrow y + y_b$ 
12:  if extract <query> </query> from  $y_b$  then
13:    Extract  $q \leftarrow \text{Parse}(y_b, \text{<query>, </query>})$ 
14:    Retrieve knowledge  $d = \mathcal{R}(q)$ 
15:    Continue rollout  $y \leftarrow y + \text{<knowledge>}d\text{</knowledge>}$ 
16:  else if extract </answer> from  $y_b$  then
17:    return  $y$ 
18:  end if
19:  count turns  $b \leftarrow b + 1$ 
20: end while
21: return  $y$ 

```

Dataset	Metric	Baseline	AdaptR1 (5 Seeds)
2WikiMultiHopQA	EM	58.59	61.72 $\pm$ 1.48
	F1	68.18	68.73 $\pm$ 1.50
	Avg. Think Tokens	151.14	25.43 $\pm$ 18.81
HotpotQA	EM	55.47	60.31 $\pm$ 1.76
	F1	63.55	65.93 $\pm$ 1.02
	Avg. Think Tokens	103.68	13.81 $\pm$ 2.22
Musique	EM	37.50	39.69 $\pm$ 1.28
	F1	48.33	50.39 $\pm$ 2.21
	Avg. Think Tokens	148.29	45.69 $\pm$ 36.88

Table 9: Robustness statistics over 5 random seeds on three representative datasets.

	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Avg. Ratio	Avg. F1
<i>Impact of Step-wise Weight (λ)</i>								
$\lambda = 0.5$	0.0000	0.0000	0.0000	0.0000	0.0000	-	0.0000	0.6827
<i>Sensitivity Analysis</i>								
$\lambda = 0.1$	0.0000	0.0000	0.5000	0.8966	0.9524	0.9524	0.4408	0.6819
$\lambda = 0.2$	0.0000	0.0000	0.3047	0.8440	0.7742	1.0000	0.3025	0.6756
$\lambda = 0.3$	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.6870
$\lambda = 0.4$	0.4219	0.4297	1.0000	1.0000	1.0000	-	0.7184	0.6599
$\lambda = 0.6$	0.0000	0.0000	0.0000	0.0000	-	-	0.0000	0.6959
$\lambda = 0.7$	1.0000	0.2344	1.0000	-	-	-	0.7216	0.6370
$\lambda = 0.8$	0.0312	0.0000	0.0000	0.0000	0.0000	-	0.0102	0.6572
$\lambda = 0.9$	1.0000	0.0000	0.9921	0.2381	0.0000	-	0.6174	0.6920

Table 10: Analysis of the No-Think ratio from step 1 to 6 and performance (F1) for Graph-R1 with and without AdaptR1 on 2WikiMultiHopQA. The parameter λ controls the step-wise penalty weight.

	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Avg. Ratio	Avg. F1
<i>Impact of Step-wise Weight (λ)</i>								
$\lambda = 0.5$	1.0000	1.0000	1.0000	-	-	-	1.0000	0.6081
<i>Sensitivity Analysis</i>								
$\lambda = 0.1$	1.0000	1.0000	-	-	-	-	1.0000	0.6272
$\lambda = 0.2$	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9972	0.6460
$\lambda = 0.3$	1.0000	1.0000	-	-	-	-	1.0000	0.6421
$\lambda = 0.4$	1.0000	1.0000	1.0000	-	-	-	1.0000	0.6109
$\lambda = 0.6$	1.0000	1.0000	1.0000	-	-	-	1.0000	0.6205
$\lambda = 0.7$	1.0000	1.0000	1.0000	-	-	-	1.0000	0.6240
$\lambda = 0.8$	1.0000	1.0000	1.0000	1.0000	1.0000	0.9167	0.9924	0.6483
$\lambda = 0.9$	1.0000	1.0000	-	-	-	-	1.0000	0.6211

Table 11: Analysis of the No-Think ratio from step 1 to 6 and performance (F1) for Graph-R1 with and without AdaptR1 on HotpotQA. The parameter λ controls the step-wise penalty weight.

	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Avg. Ratio	Avg. F1
<i>Impact of Step-wise Weight (λ)</i>								
$\lambda = 0.5$	1.0000	1.0000	1.0000	1.0000	-	-	1.0000	0.4686
<i>Sensitivity Analysis</i>								
$\lambda = 0.1$	0.9922	0.0156	0.0000	0.0000	0.0000	-	0.4868	0.4704
$\lambda = 0.2$	1.0000	1.0000	-	-	-	-	1.0000	0.4641
$\lambda = 0.3$	1.0000	1.0000	-	-	-	-	1.0000	0.4623
$\lambda = 0.4$	1.0000	1.0000	1.0000	-	-	-	1.0000	0.4875
$\lambda = 0.6$	1.0000	1.0000	-	-	-	-	1.0000	0.4842
$\lambda = 0.7$	1.0000	1.0000	-	-	-	-	1.0000	0.4937
$\lambda = 0.8$	1.0000	1.0000	-	-	-	-	1.0000	0.4729
$\lambda = 0.9$	1.0000	1.0000	-	-	-	-	1.0000	0.4914

Table 12: Analysis of the No-Think ratio from step 1 to 6 and performance (F1) for Graph-R1 with and without AdaptR1 on NQ. The parameter λ controls the step-wise penalty weight.

	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Avg. Ratio	Avg. F1
<i>Impact of Step-wise Weight (λ)</i>								
$\lambda = 0.5$	1.0000	1.0000	-	-	-	-	1.0000	0.5573
<i>Sensitivity Analysis</i>								
$\lambda = 0.1$	1.0000	1.0000	-	-	-	-	1.0000	0.5584
$\lambda = 0.2$	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.5495
$\lambda = 0.3$	1.0000	1.0000	-	-	-	-	1.0000	0.5536
$\lambda = 0.4$	1.0000	1.0000	-	-	-	-	1.0000	0.5573
$\lambda = 0.6$	1.0000	1.0000	-	-	-	-	1.0000	0.5589
$\lambda = 0.7$	1.0000	1.0000	1.0000	1.0000	1.0000	-	1.0000	0.5539
$\lambda = 0.8$	1.0000	1.0000	-	-	-	-	1.0000	0.5683
$\lambda = 0.9$	1.0000	1.0000	-	-	-	-	1.0000	0.5284

Table 13: Analysis of the No-Think ratio from step 1 to 6 and performance (F1) for Graph-R1 with and without AdaptR1 on PopQA. The parameter λ controls the step-wise penalty weight.

	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Avg. Ratio	Avg. F1
<i>Impact of Step-wise Weight (λ)</i>								
$\lambda = 0.5$	1.0000	1.0000	-	-	-	-	1.0000	0.7319
<i>Sensitivity Analysis</i>								
$\lambda = 0.1$	1.0000	1.0000	-	-	-	-	1.0000	0.7157
$\lambda = 0.2$	0.9766	1.0000	-	-	-	-	0.9883	0.7290
$\lambda = 0.3$	1.0000	1.0000	1.0000	-	-	-	1.0000	0.7156
$\lambda = 0.4$	1.0000	1.0000	-	-	-	-	1.0000	0.7279
$\lambda = 0.6$	1.0000	1.0000	-	-	-	-	1.0000	0.7213
$\lambda = 0.7$	1.0000	1.0000	-	-	-	-	1.0000	0.7100
$\lambda = 0.8$	1.0000	0.9531	-	-	-	-	0.9766	0.7181
$\lambda = 0.9$	1.0000	0.9922	-	-	-	-	0.9961	0.7277

Table 14: Analysis of the No-Think ratio from step 1 to 6 and performance (F1) for Graph-R1 with and without AdaptR1 on TriviaQA. The parameter λ controls the step-wise penalty weight.

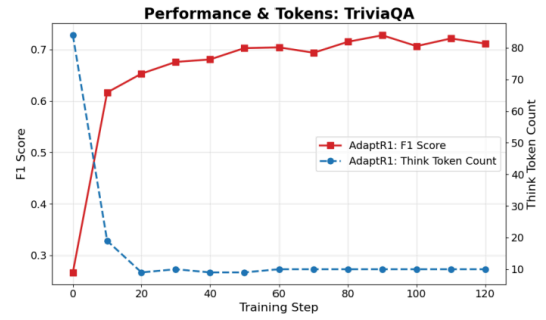
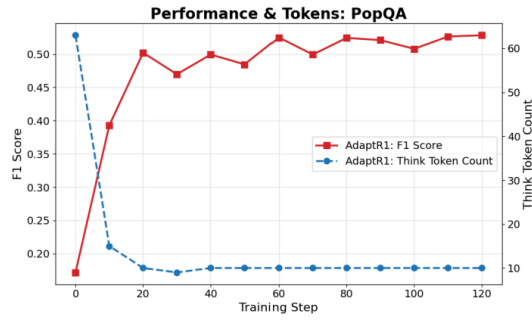
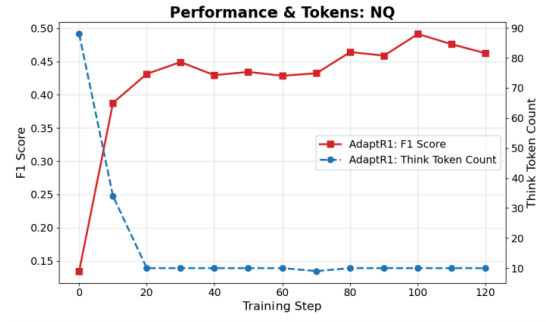
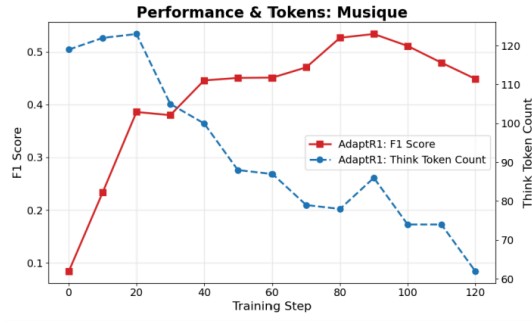
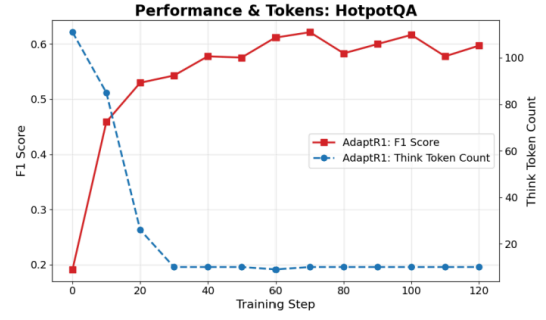
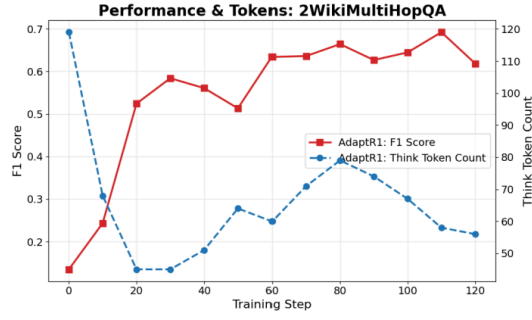


Figure 4: Think tokens and F1 scores in the training steps for six datasets.

E.5 Failure Analysis

We analyze when `no_think` can hurt answer quality by comparing Graph-AdaptR1 with the full-thinking Graph-R1 baseline. Table 17 separates examples that only AdaptR1 answers correctly, examples both methods answer correctly, and cases where Graph-R1 is correct but Graph-AdaptR1 fails. The last category approximates failures where explicit reasoning was likely useful but was skipped or shortened too aggressively.

The failures are relatively infrequent, and in all datasets except NQ the number of examples fixed by AdaptR1 is larger than the number lost by AdaptR1. Qualitatively, the main failure mode appears in strict comparison questions that require precise extraction and comparison of attributes, such as dates or numerical values. For example, for the question “Which film has the director who died later, *The Hellions* or *Hum Kaun Hai*?” Graph-AdaptR1 answers *The Hellions* after skipping final synthesis, while Graph-R1 retrieves the directors separately, compares their death years, and answers *Hum Kaun Hai*. This suggests that `no_think` is most risky when the final step requires explicit symbolic comparison over retrieved evidence.

F Experimental Settings

F.1 Baselines in the Graph-R1 Setting

We categorize our baselines into two distinct groups based on the underlying backbone model.

GPT-4o-mini Based Methods. The first group employs **GPT-4o-mini** as an inference-only generator. We evaluate the base model’s intrinsic capacity using **NaiveGeneration**, a zero-shot approach without retrieval. We also include **Standard-RAG** (Lewis et al., 2020), representing the conventional chunk-based retrieval-augmented generation paradigm. Furthermore, we assess a suite of graph-based retrieval strategies: **GraphRAG** (Edge et al., 2025), which constructs entity graphs for one-shot retrieval; **LightRAG** (Guo et al., 2025), a streamlined variant designing compact graphs for efficiency; **PathRAG** (Chen et al., 2025a), which executes retrieval via path-based pruning on entity graphs; **HippoRAG2** (Gutiérrez et al., 2025), utilizing a hierarchical path planner over knowledge graphs; and **HyperGraphRAG** (Luo et al., 2025b), which leverages n-ary relational hypergraphs to facilitate single-step retrieval.

Qwen2.5-Instruct Based Methods. The second group utilizes the **Qwen2.5-Instruct (7B)** model.

We establish foundational performance bounds using **NaiveGeneration**, the classic **StandardRAG** (Lewis et al., 2020) pipeline, and **SFT** (Zheng et al., 2024), which applies supervised fine-tuning on QA pairs. Additionally, we evaluate advanced methods optimized via reinforcement learning (RL): **R1** (Shao et al., 2024), a policy trained with GRPO to generate answers directly without retrieval; **Search-R1** (Jin et al., 2025a), a multi-turn chunk-based retrieval approach trained via GRPO; **R1-Searcher** (Song et al., 2025), a two-stage GRPO-based framework for chunk-based retrieval; and **Graph-R1** (Luo et al., 2025a), an agentic GraphRAG framework enhanced by end-to-end reinforcement learning.

F.2 Evaluation Metrics

We assess model performance using three primary metrics focusing on answer accuracy and retrieval quality.

Exact Match (EM). We employ Exact Match to strictly evaluate answer accuracy. This metric determines whether the generated answer y_i is identical to the ground-truth reference y_i^* following a normalization process (i.e., lowercasing, punctuation removal, and whitespace standardization). The EM score is averaged over all N samples:

$$\text{EM} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[\text{norm}(y_i) = \text{norm}(y_i^*)], \quad (11)$$

where $\mathbb{I}[\cdot]$ denotes the indicator function.

F1 Score. To provide a more granular assessment of generation quality beyond binary matching, we utilize the F1 score. This metric measures the token-level overlap between the prediction and the ground truth, defined as the harmonic mean of precision and recall:

$$\text{F1} = \frac{1}{N} \sum_{i=1}^N \frac{2 \cdot |\mathcal{T}(y_i) \cap \mathcal{T}(y_i^*)|}{|\mathcal{T}(y_i)| + |\mathcal{T}(y_i^*)|}, \quad (12)$$

where $\mathcal{T}(\cdot)$ represents the set of tokens in a given text.

Retrieval Similarity (R-S). To assess the efficacy of the retrieval module in isolation, we compute Retrieval Similarity. This metric measures the semantic alignment between the retrieved context $k_{\text{retr}}^{(i)}$ and the ground-truth “gold” context $k_{\text{gold}}^{(i)}$. We utilize a semantic embedding function, $\text{Enc}(\cdot)$, to compute the cosine similarity between the vector

representations:

$$R-S = \frac{1}{N} \sum_{i=1}^N \cos \left(\text{Enc}(k_{\text{retr}}^{(i)}), \text{Enc}(k_{\text{gold}}^{(i)}) \right). \quad (13)$$

F.3 Implementation Details

We outline the general hyperparameters in Table 18. While our AdaptR1 models largely inherit the configurations from their respective backbones, we introduce specific adjustments to the reinforcement learning setup to facilitate adaptive training. We configure the group size G (number of rollouts per query) to 8 for Graph-AdaptR1 and 5 for Search-AdaptR1. For Search-AdaptR1, we add the step-wise No-Think reward to the original answer and retrieval set rewards. For the AdaptR1-specific coefficients, we set the confidence threshold $\tau = 0.6$, the reward weight $\omega = 0.2$, and the step-wise penalty factor $\lambda = 0.9$. Main and ablation experiments were conducted on NVIDIA H200 GPUs; robustness statistics in Table 9 are repeated over 5 random seeds on three representative datasets. The total computational budget for all reported experiments (across all datasets and ablation studies) was approximately 1500 GPU hours.

G Details of Research Artifacts and Licenses

G.1 Datasets and Licenses

In this work, we utilize six publicly available datasets to evaluate the multi-hop reasoning capabilities of our model. All datasets are widely used in the research community, and our use is consistent with their intended use for research and evaluation purposes.

- **2WikiMultiHopQA** (Ho et al., 2020): A multi-hop QA dataset using structured and unstructured data. It is distributed under the **Apache-2.0 License**.
- **HotpotQA** (Yang et al., 2018): A dataset with question-answer pairs based on Wikipedia articles, distributed under the **CC BY-SA 4.0 License**.
- **MuSiQue** (Trivedi et al., 2022b): A dataset for multi-hop reasoning over connected paragraphs, distributed under the **CC BY 4.0 License**.
- **Natural Questions (NQ)** (Kwiatkowski et al., 2019): A dataset consisting of queries issued to the Google search engine, distributed under the **Apache-2.0 License**.
- **PopQA** (Mallen et al., 2023): A dataset focusing on long-tail knowledge retrieval using entity-centric questions, distributed under the **MIT License**.
- **TriviaQA** (Joshi et al., 2017): A reading comprehension dataset containing question-answer-evidence triples, distributed under the **Apache-2.0 License**.

These datasets are primarily in English and are derived from public sources such as Wikipedia or Web snippets. Aligned with the experimental setup of Graph-R1 (Luo et al., 2025a), we standardize our data usage by uniformly sampling 5,120 instances for training and 128 instances for testing per dataset, thereby balancing computational workload and consistency.

G.2 Models, Frameworks and their licenses

We conduct our training and evaluation using the following models and frameworks:

- **Language Model:**

We use **Qwen2.5-7B-Instruct** (Qwen et al., 2025) as our backbone model. Qwen2.5 is open-sourced under the **Apache-2.0 License**, allowing for research and commercial use.

- **Retrievers:**

The choice of retriever depends on the specific method employed. In Search-R1, we utilize **E5** (Wang et al., 2022). In Graph-R1, we employ hypergraph-based retrieval equipped with **bge-large-en-v1.5** (Chen et al., 2023). Both embedding models are open-sourced under the **MIT License**.

- **Framework:**

We implement our methods using **VeRL** (Sheng et al., 2025), a flexible framework for reinforcement learning with LLMs. The VeRL library is open-sourced under the **Apache-2.0 License**.

Question: What is the birthplace of the man who does the voice of Stan on the series that includes the episode The Hobbit?

Ground Truth: "Denver" or "Denver, Colorado"

Graph-R1:

<think> To answer this question, I need to find the birthplace of the voice actor who does the voice of Stan in the series that includes the episode "The Hobbit". First, I will determine the name of the series that includes the episode "The Hobbit". Then, I will find the birthplace of the actor who voices Stan in that series.

</think>

<query> {"query": "what series includes the episode The Hobbit?"} **</query>**

<knowledge> {"results": [{"<knowledge>": "The Hobbit is the tenth episode in the seventeenth season of 'South Park', with a storyline centered on Wendy Testaburger's attempts at body image awareness.", "<coherence>": 2.333, "<knowledge>": "The Hobbit is the tenth and final episode in the seventeenth season of the American animated television series South Park. The 247th episode of the series overall", "<coherence>": 1.283, "<knowledge>": "It was commissioned for the soundtrack of the 2013 film 'The Hobbit', where it was played over the closing credits.", "<coherence>": 0.667, "<knowledge>": "Kidnapped is a two-part BBC television adaptation of the 1886 novel of the same name by Robert Louis Stevenson.", "<coherence>": 0.25, "<knowledge>": "'The Hobbit' director, Peter Jackson, asked Sheeran to write a song for the movie after Jackson's daughter, Katie, suggested Sheeran.", "<coherence>": 0.2, "<knowledge>": "The episode received positive reviews, praising Wendy's story arc and the return of Kanye West.", "<coherence>": 0.143, "<knowledge>": "The episode is noted as the 247th episode of 'South Park', showcasing the series' long run and significance.", "<coherence>": 0.125}]} **</knowledge>**

<think> Based on the knowledge base, the series that includes the episode "The Hobbit" is South Park. Now, I need to find the birthplace of the actor who voices Stan in South Park. **</think>**

<query> {"query": "birthplace of the actor who voices Stan in South Park"} **</query>**

<knowledge> {"results": [{"<knowledge>": "Trey Parker voices several main characters including Stan Marsh, Eric Cartman, Randy Marsh, and Mr. Garrison, as well as recurring characters in 'South Park.'", "<coherence>": 3.283, "<knowledge>": "Dian Bachar (; born October 26, 1970, in Denver, Colorado) is an American actor most notable for his roles in various films by or starring his friends Trey Parker and Matt Stone, such as Cannibal! The Musical (George Noon), Orgazmo (Ben Chapleski) and his most famous role as Kenny Squeak Scolari in 1998's BASEketball, as well as making the occasional appearance on South Park.", "<coherence>": 0.5, "<knowledge>": "Big Stan is a 2007 American prison comedy directed and produced by Rob Schneider, who also starred in the film.", "<coherence>": 0.333, "<knowledge>": "Kenny 'Squeak' Scolari was known for his role in 1998's 'BASEketball', along with appearances on 'South Park' and portraying an alien engineer in 'Galaxy Quest'.", "<coherence>": 0.25, "<knowledge>": "He was raised in Collingwood, Ontario.", "<coherence>": 0.2}]} **</knowledge>**

<think> Based on the knowledge base, Trey Parker voices Stan Marsh in South Park. Therefore, I need to find the birthplace of Trey Parker. **</think>**

<query> {"query": "birthplace of Trey Parker"} **</query>**

<knowledge> {"results": [{"<knowledge>": "Henry Benbridge (October 1743 u2013 February 1812) was an early American portrait painter.", "<coherence>": 1.0}, {"<knowledge>": "Born in Enugu, Nigeria on March 1, 1980, Anthony Njokuani and his family immigrated to the United States in 1983, settling in Garland, Texas.", "<coherence>": 1.0}, {"<knowledge>": "Dian Bachar (; born October 26, 1970, in Denver, Colorado) is an American actor most notable for his roles in various films by or starring his friends Trey Parker and Matt Stone, such as Cannibal! The Musical (George Noon), Orgazmo (Ben Chapleski) and his most famous role as Kenny Squeak Scolari in 1998's BASEketball, as well as making the occasional appearance on South Park.", "<coherence>": 0.553}, {"<knowledge>": "u201cAre You Ready for Love" is a song recorded by Elton John in 1977 and first released in the UK in 1979 on the EP "The Thom Bell Sessions". It was written by Leroy Bell, Thom Bell and Casey James, and was originally produced in Philadelphia by Thom Bell.", "<coherence>": 0.5}, {"<knowledge>": "Trey Parker voices several main characters including Stan Marsh, Eric Cartman, Randy Marsh, and Mr. Garrison, as well as recurring characters in 'South Park.'", "<coherence>": 0.383}, {"<knowledge>": "The Federal Detention Center (FDC Philadelphia) is a United States Federal prison in Center City, Philadelphia, Pennsylvania.", "<coherence>": 0.333}, {"<knowledge>": "The development of the appreciation medals was initiated by the White House and executed by the Bureau of the Mint through the U.S. Mint in Philadelphia.", "<coherence>": 0.25}, {"<knowledge>": "United States. It was the birthplace and boyhood home of Dr. George Horace Gallup, the pioneer of public opinion polling.", "<coherence>": 0.25}, {"<knowledge>": "He was born in Philadelphia, Pennsylvania, the eldest son of a well-known Quaker family (Richard and Anne Roberts Vaux) and connected by marriage to another such family, the Wistars.", "<coherence>": 0.2}, {"<knowledge>": "Nathaniel Carlyle Adderley was born in Tampa, Florida, but moved to Tallahassee when his parents were hired to teach at Florida A&M University.", "<coherence>": 0.2}]}]} **</knowledge>**

<think> Based on the knowledge base, Trey Parker was born in Philadelphia, Pennsylvania. **</think>**

<answer> Philadelphia, Pennsylvania **</answer>**

Table 15: A case study of Graph-R1.

Question: What is the birthplace of the man who does the voice of Stan on the series that includes the episode The Hobbit?

Ground Truth: "Denver" or "Denver, Colorado"

Graph-AdaptR1:

<think> no_think </think>

<query> {"query": "What is the birthplace of the man who does the voice of Stan on the series that includes the episode The Hobbit?"} </query>

<knowledge> {"results": [{"<knowledge>": "Trey Parker voices several main characters including Stan Marsh, Eric Cartman, Randy Marsh, and Mr. Garrison, as well as recurring characters in 'South Park.'", "<coherence>": 1.5}, {"<knowledge>": "with 'Stan Cohen'."<coherence>": 1.0}, {"<knowledge>": "The film co-stars Jennifer Morrison, Scott Wilson and David Carradine.", "<coherence>": 0.583}, {"<knowledge>": "The Lorax (also known as Dr. Seuss' The Lorax) is a 2012 American 3D computer-animated musical fantasy-comedy film produced by Illumination Entertainment and based on Dr. Seuss's children's book of the same name.", "<coherence>": 0.5}, {"<knowledge>": "The Hobbit is the tenth and final episode in the seventeenth season of the American animated television series South Park. The 247th episode of the series overall", "<coherence>": 0.4}, {"<knowledge>": "The Hobbit is the tenth episode in the seventeenth season of 'South Park', with a storyline centered on Wendy Testaburger's attempts at body image awareness.", "<coherence>": 0.333}, {"<knowledge>": "Dian Bachar (; born October 26, 1970, in Denver, Colorado) is an American actor most notable for his roles in various films by or starring his friends Trey Parker and Matt Stone, such as Cannibal! The Musical (George Noon), Orgazmo (Ben Chapleski) and his most famous role as Kenny Squeak Scolari in 1998's BASEketball, as well as making the occasional appearance on South Park.", "<coherence>": 0.25}]} </knowledge>

<think> The man who does the voice of Stan on the series that includes the episode "The Hobbit" is Trey Parker. The birthplace of Trey Parker is Denver, Colorado. </think>

<query> {"query": "What is the birthplace of Trey Parker?"} </query>

<knowledge> {"results": [{"<knowledge>": "Henry Benbridge (October 1743 u2013 February 1812) was an early American portrait painter.", "<coherence>": 1.0}, {"<knowledge>": "Trey Parker voices several main characters including Stan Marsh, Eric Cartman, Randy Marsh, and Mr. Garrison, as well as recurring characters in 'South Park.'", "<coherence>": 1.0}, {"<knowledge>": "u201cAre You Ready for Love" is a song recorded by Elton John in 1977 and first released in the UK in 1979 on the EP "The Thom Bell Sessions". It was written by Leroy Bell, Thom Bell and Casey James, and was originally produced in Philadelphia by Thom Bell.", "<coherence>": 0.5}, {"<knowledge>": "Dian Bachar (; born October 26, 1970, in Denver, Colorado) is an American actor most notable for his roles in various films by or starring his friends Trey Parker and Matt Stone, such as Cannibal! The Musical (George Noon), Orgazmo (Ben Chapleski) and his most famous role as Kenny Squeak Scolari in 1998's BASEketball, as well as making the occasional appearance on South Park.", "<coherence>": 0.5}, {"<knowledge>": "The Federal Detention Center (FDC Philadelphia) is a United States Federal prison in Center City, Philadelphia, Pennsylvania.", "<coherence>": 0.333}, {"<knowledge>": "United States. It was the birthplace and boyhood home of Dr. George Horace Gallup, the pioneer of public opinion polling.", "<coherence>": 0.333}, {"<knowledge>": "The development of the appreciation medals was initiated by the White House and executed by the Bureau of the Mint through the U.S. Mint in Philadelphia.", "<coherence>": 0.25}, {"<knowledge>": "Nathaniel Carlyle Adderley was born in Tampa, Florida, but moved to Tallahassee when his parents were hired to teach at Florida A&M University.", "<coherence>": 0.25}, {"<knowledge>": "He was born in Philadelphia, Pennsylvania, the eldest son of a well-known Quaker family (Richard and Anne Roberts Vaux) and connected by marriage to another such family, the Wistars.", "<coherence>": 0.2}, {"<knowledge>": "Cassavetes was born in New York City, the son of Greek American Katherine Cassavetes and Greek immigrant Nicholas John Cassavetes.", "<coherence>": 0.2}]} </knowledge>

<think> no_think </think>

<answer> Denver, Colorado </answer>

Table 16: A case study of Graph-AdaptR1.

Dataset	Only AdaptR1 Correct	Both Correct	Only Graph-R1 Correct
2WikiMultiHopQA	15	64	11
HotpotQA	14	60	11
Musique	13	38	10
NQ	3	43	2
PopQA	7	56	9
TriviaQA	4	78	7

Table 17: Overlap analysis between Graph-AdaptR1 and Graph-R1. "Only Graph-R1 Correct" indicates cases where skipping or shortening explicit reasoning likely harms the answer.

Method	Backbone	Batch Size	Max Length	Top-K	Algo	Epochs
NaiveGeneration	Qwen2.5 / GPT-4o-mini	–	∞	N/A	–	–
StandardRAG	Qwen2.5 / GPT-4o-mini	–	∞	5 Chunks	–	–
GraphRAG	GPT-4o-mini	–	∞	60	–	–
LightRAG	GPT-4o-mini	–	∞	60	–	–
PathRAG	GPT-4o-mini	–	∞	60	–	–
HippoRAG2	GPT-4o-mini	–	∞	60	–	–
HyperGraphRAG	GPT-4o-mini	–	∞	60	–	–
SFT	Qwen2.5 (7B)	16	4096	N/A	LoRA	3
R1	Qwen2.5 (7B)	128	4096	N/A	GRPO	3
R1-Searcher	Qwen2.5 (7B)	128	4096	5 Chunks / Turn	GRPO	3
Search-R1	Qwen2.5 (7B)	128	4096	5 Chunks / Turn	GRPO	6
Search-AdaptR1	Qwen2.5 (7B)	128	4096	5 Chunks / Turn	GRPO	6
Graph-R1	Qwen2.5 (7B)	128	4096	5 Chunks / Turn	GRPO	3
Graph-AdaptR1	Qwen2.5 (7B)	128	4096	5 Chunks / Turn	GRPO	3

Table 18: Hyperparameter settings in Graph-R1 setting.